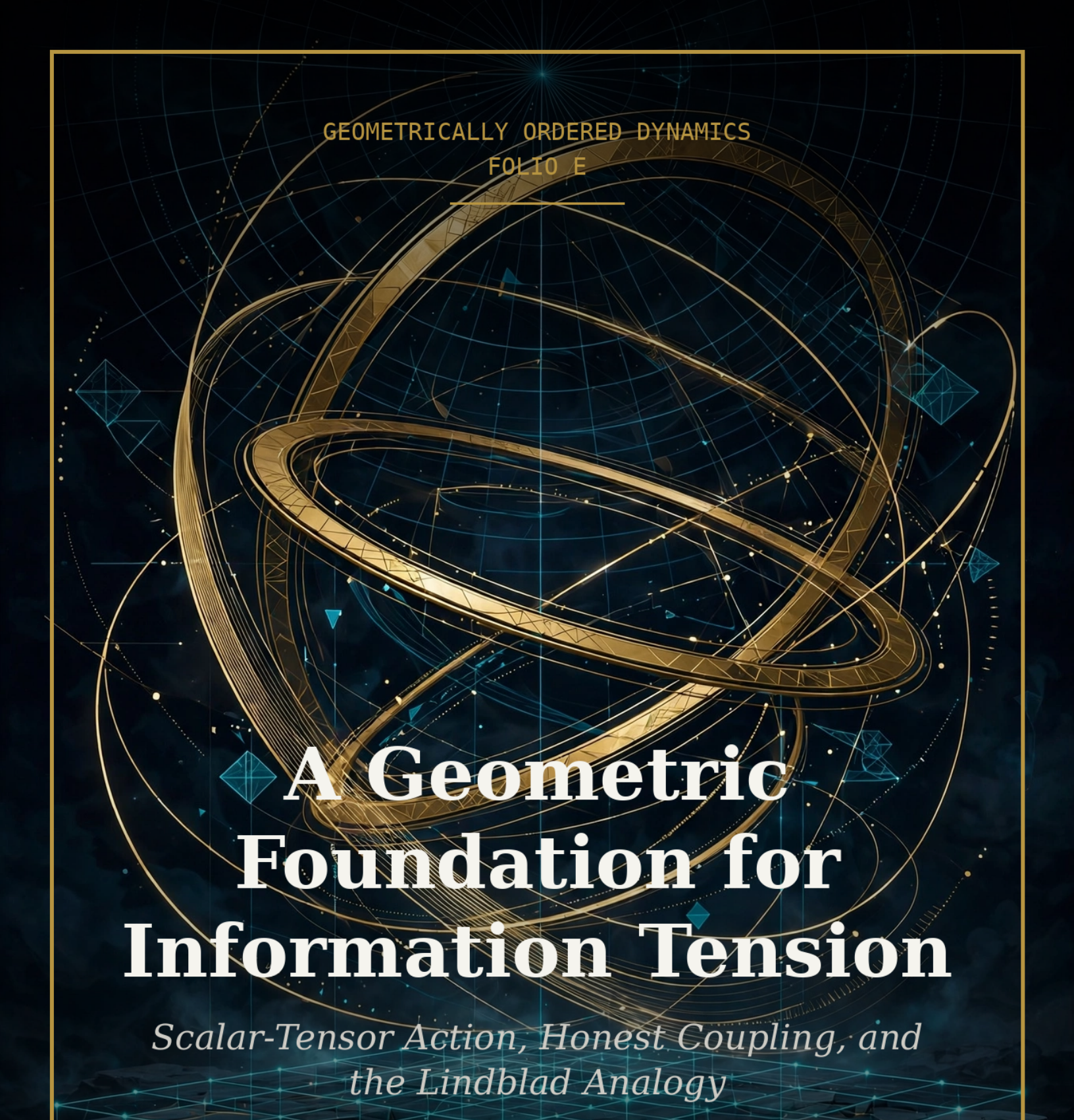
GEOMETRICALLY ORDERED DYNAMICS
FOLIO E



A Geometric Foundation for Information Tension

Scalar-Tensor Action, Honest Coupling, and the Lindblad Analogy

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RESULTS STATED IN TIER — LIMITS FIRST

CONVERGENCE -- INFORMATION TENSION

This folio converges on **information tension** from the **geometric-foundations** angle: the scalar-tensor action it is derived from, the honest statement of its non-minimal coupling, and the Lindblad connection stated as what it is — an analogy. Sibling papers approach the same claim from other angles: [TIER: CONJECTURE] the geometric-phase-force successor (IT-FIELD companion) carries the force-mechanism angle — the Sovereign Action, the tension tensor as force, and the horizon acceleration scale — and the Holonomery successor (IT-MARKET) carries the observer-substrate angle of the same parallel-transport mathematics. Limits first: this folio contains no empirical dataset; its numbers are fits and modeling choices, and the one sentence that claimed otherwise has been struck.

SUPERSESSION NOTE

This folio **supersedes** Zenodo record 20652219, *A Geometric Foundation for Information Tension and Galactic Dynamics*, per the canon rule: the correction *becomes* the paper. Named repairs: (1) the claim that the fitted $\xi \approx 0.127$ is "well within" the Cassini bound $|\xi| < 10^{-3}$ was **backwards** — 0.127 exceeds the bound by a factor of ∼130 — and the consistency claim is retracted (§2); (2) the abstract's "this unification is not metaphorical" contradicts the paper's own Limitations admitting the Lindblad connection "is an analogy" — the honest sentence is kept: **it is an analogy** (§5); (3) the Trinity 2.0 empirical section is removed: the dataset was RNG-generated synthetic data rescaled to the target, retracted corpus-wide; (4) $\kappa = 0.9539$ "emerges naturally" is reworded to what the paper's own Eq. 11 does: it is **fit** from $(\xi, \lambda/m^2)$; (5) the Eq. 13–14 arithmetic is corrected ($\kappa^2/(1-\kappa^2) \approx 10.1$, not 11.1; the 12.62 multiplier was unexplained and is struck).

1 The GOD action and the information tension tensor

The loadbearing structure is the scalar-tensor action for the sovereign scalar field Φ on a pseudo-Riemannian spacetime $(M, g_{\mu\nu})$ — a legitimate modified-gravity framework shape, independent of the numbers the old record attached to it:

$$S_{\mathrm{GOD}} = \int_M \left[\frac{R}{16\pi G} + \frac{1}{2} g^{\mu\nu} \partial_\mu \Phi \, \partial_\nu \Phi - V(\Phi) + \xi R \Phi^2 \right] \sqrt{-g}\, d^4x, \tag{1}$$

with $V(\Phi) = \frac{1}{2} m^2 \Phi^2 + \frac{\lambda}{4} \Phi^4$. Varying with respect to $g_{\mu\nu}$ gives $G_{\mu\nu} + \kappa T_{\mu\nu} = 0$, where $T_{\mu\nu}$ is the information tension tensor: the stress-energy of Φ, measuring the local flux of topological order [TIER: CONJECTURE]. Varying with respect to Φ gives the field equation $\nabla^\mu \nabla_\mu \Phi - m^2 \Phi - \lambda \Phi^3 + 2\xi R \Phi = 0$. This section is **mathematics**, not measurement.

2 The coupling, stated honestly

The superseded record fitted the non-minimal coupling as $\xi \approx 0.127$ (with $\lambda/m^2 \approx 0.84$) and then claimed this was "well within the Cassini bound $|\xi| < 10^{-3}$." That sentence was **wrong in direction**: 0.127 exceeds 10^{-3} by a factor of roughly **130**. The old record's "screening mechanisms" hand-wave

does not rescue it — and the old record's own Limitations section admitted the screening bound "has not been verified rigorously in the non-linear regime."

The honest statement, which replaces it: the value $\xi \approx 0.127$ is a **fit**, and as fitted it is **in tension with the Cassini bound**, not within it. Either a screening derivation must be produced or the consistency claim stays retracted. Both are open work; neither is done in this folio. This correction is the single most important repair in the record.

3 Modified cosmology and the corrected arithmetic

In flat FLRW the 00-component gives the modified Friedmann equation

$$H^2 = \frac{8\pi G}{3}(\rho_m + \rho_\Lambda + \rho_T), \tag{2}$$

with the tension density satisfying the self-consistency relation

$$\rho_T = \frac{\kappa^2}{1-\kappa^2}(\rho_m + \rho_\Lambda). \tag{3}$$

The old record printed $\rho_T \approx 11.1(\rho_m+\rho_\Lambda)$; the arithmetic is $\kappa^2/(1-\kappa^2) = 0.9099/0.0901 \approx \mathbf{10.1}$. The 11.1 is struck. The old record's Eq. 14 further multiplied by an **unexplained 12.62** to manufacture $\bar{\Lambda} \approx 140.1$ "in agreement" with the Trinity mean — that multiplier had no derivation, and the Trinity number it agreed with was synthetic. Both are struck. The flux-boost formula is retired with the dataset it was tuned to.

4 The sovereign invariant: a fit, not an emergence

The old abstract stated that $\kappa = 0.9539$ "emerges naturally as the critical damping coefficient of the vacuum." The old paper's own derivation (its Eq. 11) does the opposite: it **fits** κ from the parameter combination $(\xi, \lambda/m^2)$. The sentence is reworded to what was actually done: $\kappa \approx 0.9539$ is a **fit parameter** of the action family, [TIER: MODEL], not a derived constant. The ± 0.0012 "empirical" uncertainty the old record quoted came from Trinity 2.0 and is struck with it. A systematic survey of alternative action families — needed to establish any uniqueness — remains open, as the old Limitations admitted.

5 The Lindblad connection is an analogy

The old abstract and introduction insisted "this unification is not metaphorical"; the old paper's own Limitations section then admitted: "The Lindblad connection in Section 4 is an analogy." One of these sentences had to go. **The honest one stays: it is an analogy.** The dictionary — sovereign field $\Phi \leftrightarrow$ cognitive state ρ, tension $T_{\mu\nu} \leftrightarrow$ decoherence superoperator, chiral bound $\chi \geq 1/\sqrt{2} \leftrightarrow$ coherence floor, sovereign invariant $\kappa \leftrightarrow$ spectral gap — is a **stated analogy** [TIER: CONJECTURE], suggestive and honestly labeled, with no exhibited functor from the GOD framework into open-quantum-system dynamics (isomorphism rule: analogy, never bare isomorphism).

Accordingly: the "Information Tension Bound" theorem and its "unique" ADCCL-threshold derivation are re-stated as **conjectured within the analogy** — the interpolation formula for $\theta = 0.7$ is numerology carried by the analogy, not a derivation from the bound. The Schott-energy / non-Markovian correction is retained as a modeling choice [TIER: CONJECTURE].

6 The SPARC program, honestly framed

The old record's one empirical program worth keeping is the a_0 SPARC-fit program [TIER: CONJECTURE]: fitting the low-acceleration scale against the SPARC rotation-curve database. It is kept here as a **program**, not a result — pending re-fit against honest numbers, and with the understanding that the horizon-scale derivation lives in the companion force-mechanism folio. No SPARC numbers are printed in this record; there is no new analysis here.

7 Predictions, restated in tier

- **Retired:** the redshift-gradient prediction $\Lambda(z) \approx 141.99 \cdot (1+z)^{\kappa}$ — it was tuned to the retracted Trinity dataset. Retired, not repaired.
- **Kept as [TIER: CONJECTURE]:** if $T_{\mu\nu}$ is a genuine tensor field, it sources a quadrupole anisotropy aligned with the CMB dipole at the 10^{-5} level in temperature and B-mode polarization — testable with Simons Observatory maps.
- **Kept as [TIER: CONJECTURE], open:** a spectral gap at $\Delta \geq \kappa^2$ in a trapped-ion SU(2) lattice gauge simulator (≥ 50 qubits) — a real spectroscopic or lattice test is the honest evidential path; until then the Yang-Mills identification is an un-discharged axiom.

8 Limitations (kept and strengthened)

The old record's Limitations section was its most honest page, and it is kept in full: the Trinity dataset was compiled and analyzed by the same group (now superseded by the retraction); the κ derivation assumes one specific action family; the Lindblad connection is an analogy (now stated everywhere, not only here); the Cassini screening bound is unverified in the non-linear regime (now joined by the honest admission that the fit *violates* the bound as stated). Added: this folio contains no empirical dataset of its own, and its corrected arithmetic (§3) has not been independently re-checked.

References. Superseded: Zenodo record 20652219. Retraction record for Trinity 2.0: the Law of G.O.D. audit ledger, carried in 10.5281/zenodo.21277228 (Ars Magna) and the Ramanujan-Yett Hamiltonian v1.1, 10.5281/zenodo.21504896. Data program: SPARC (Lelli, McGaugh & Schombert 2016, *AJ* **152**, 157). Theory references as in the superseded record: Cover & Thomas (2006); Planck Collaboration (2020, *A&A* **641**, A6); Connes (1994); Jaffe & Witten (Clay Mathematics Institute, 2000); Shannon (1948).